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Transmission of AI Super Surplus in Korea - Mostly Through FX and Fiscal Policy

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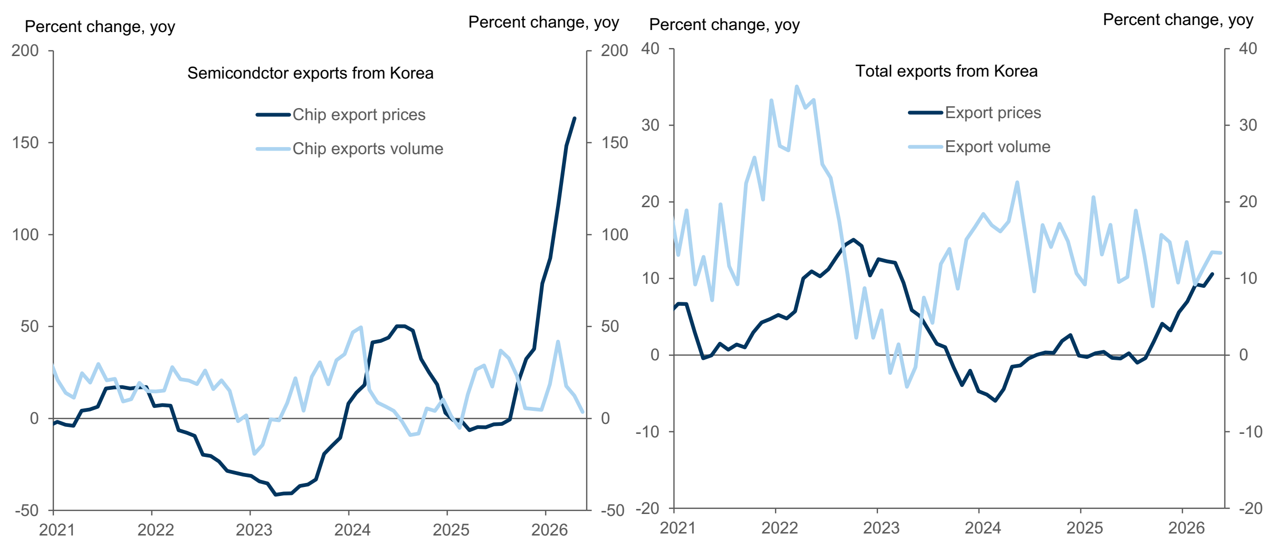
- Korea is among the major beneficiaries of the AI boom. Exports have surged on sharp memory-price gains and are on track to surpass US$1 trillion in 2026. We use cross-country terms-of-trade (TOT) case studies to assess the macro implications of large AI-driven TOT gains in Korea.
- Our cross-country study shows that positive TOT shocks are generally associated with currency appreciation across a broad set of exporters. However, the broad macro effects differ meaningfully between commodity and manufacturing exporters. Commodity exporters tend to experience stronger consumption and inflation but weaker capex, while manufacturing exporters tend to see more stable consumption, lower inflation, and stronger capex. The key distinction lies in how export-income gains are transmitted to consumption and investment across these export groups.
- For Korea, whether large TOT gains translate into broader growth will likely depend on FX and fiscal policy. The KRW has recently weakened despite large TOT gains, contrary to the cross-country stylized facts. Given the strong FX passthrough to inflation in Korea, the drag from a weaker KRW on consumers' purchasing power and domestic firms' profit margins, and the diminished export boost from KRW depreciation due to structural changes, sustained KRW weakness could offset some positive TOT spillovers to domestic demand. Fiscal policy, including the implementation of AI mega projects, is also an important factor shaping the timing and scale of TOT spillovers. The newly announced AI projects, if implemented in line with previous initiatives for creating semiconductor clusters, could boost annual real GDP growth by 0.4pp over the medium term.
- Taken together, KRW stability would need to be a near-term policy priority to secure price stability and extend the AI boom to domestic demand. Sustained KRW weakness at the outset of the AI boom cycle would add cost-push inflation, weigh on consumption, and widen the divergence between export and domestic industries. Over time, however, the policy challenge is likely to shift from stabilizing prices toward recycling Korea's external surplus productively. Beyond the near term, a mix of gradual monetary tightening and saving part of the fiscal windfall via a new sovereign wealth fund would help preserve macro stability and broaden AI-related gains across sectors. If the AI boom persists, surplus recycling will become increasingly important for both macro stability and potential

growth.

Korea is among the major beneficiaries of the AI boom. Korea's exports have recently surged and are on track to surpass US$1 trillion in 2026. This strong performance is mostly due to sharp gains in export prices, especially for memory chips, in contrast to Taiwan, where volume growth dominated export gains. Korea's memory exports rose nearly 150% from a year ago in the first five months of 2026, with nearly 80% of the rise coming from price jumps (Exhibit 1). Surging semiconductor exports—equivalent to a quarter of Korea's total exports in 2025—contributed three quarters of total year-on-year export gains through May. Overall, export price increases accounted for nearly half of total export gains over the same period, with the share exceeding half in high-frequency data.

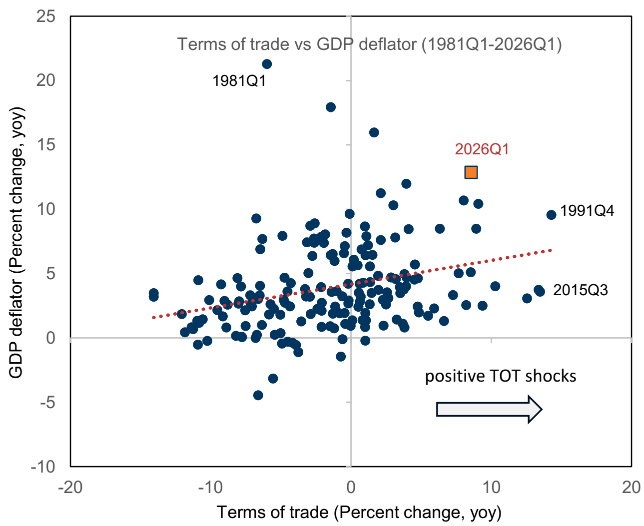
Exhibit 1: Memory price gains have driven most of Korea's recent export increase

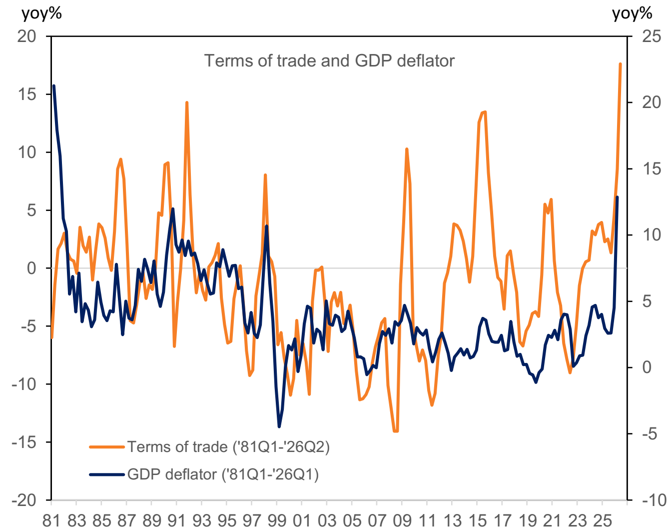


Source: Haver Analytics

These export-price gains matter beyond trade data because they are already feeding through to Korea's national-income measures. Korea's terms-of-trade (TOT; the ratio of export prices to import prices) improved 8.6% in Q1 from a year ago and accelerated further about 20% in Q2, mainly driving the roughly 13% increase in the overall price level of GDP—the GDP deflator—in Q1 (Exhibit 2). Together with 3.8% real GDP growth, the double-digit GDP deflator increase lifted nominal GDP 17.1% yoy in Q1, the fastest pace in three decades.

Exhibit 2: Korea's overall terms of trade have improved sharply, pushing up its GDP deflator





Source: Haver Analytics, Goldman Sachs Global Investment Research

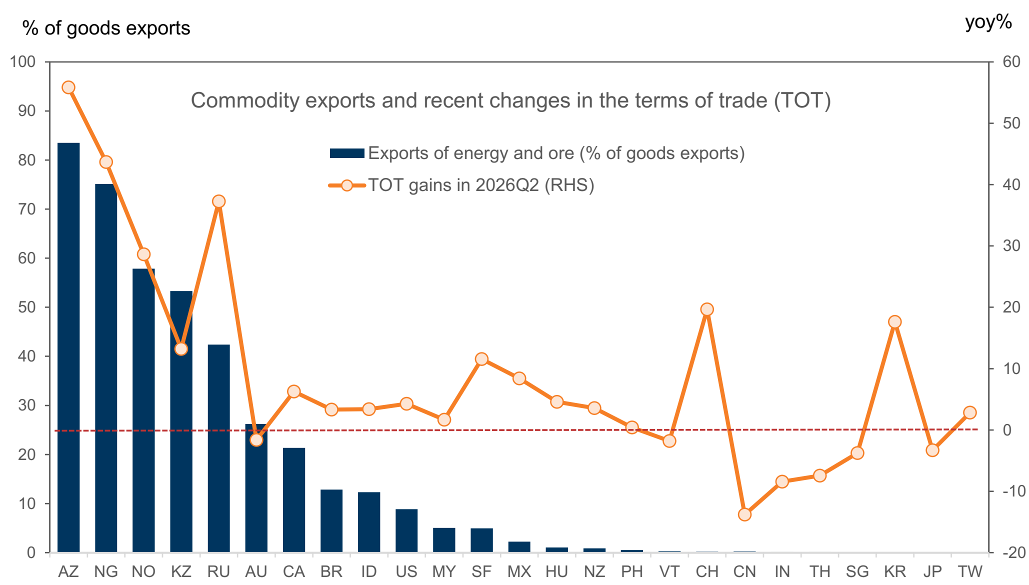
Macro Impact of Terms-of-trade Shocks—Country Cases

To gauge how such a large price-led export shock could transmit through the broader economy, we undertake case studies of the transmission of terms-of-trade (TOT) shocks to assess the macro implications of massive AI-driven TOT gains in Korea.

We examine how terms-of-trade shocks map into inflation, private consumption, and investment between 2000Q1 and 2026Q1. Our sample includes 32 major economies in our coverage, excluding the US and economies with exchange-rate pegs in the Middle East. Within the sample, we distinguish between exporters of extractive commodities—energy and ores—and other economies, since commodity exporters tend to experience terms-of-trade volatility on a scale comparable to Korea's. The terms of trade improved by nearly 20% in Korea in Q2, comparable to those of many commodity exporters such as Chile, South Africa, Kazakhstan, Brazil, and Indonesia (Exhibit 3). Extractive commodity sectors also require large capex, similar to the memory sector.[1]

[1] The exporters of extractive commodities include eight economies: Australia, Brazil, Chile, Indonesia, Kazakhstan, Norway, Russia, and South Africa.

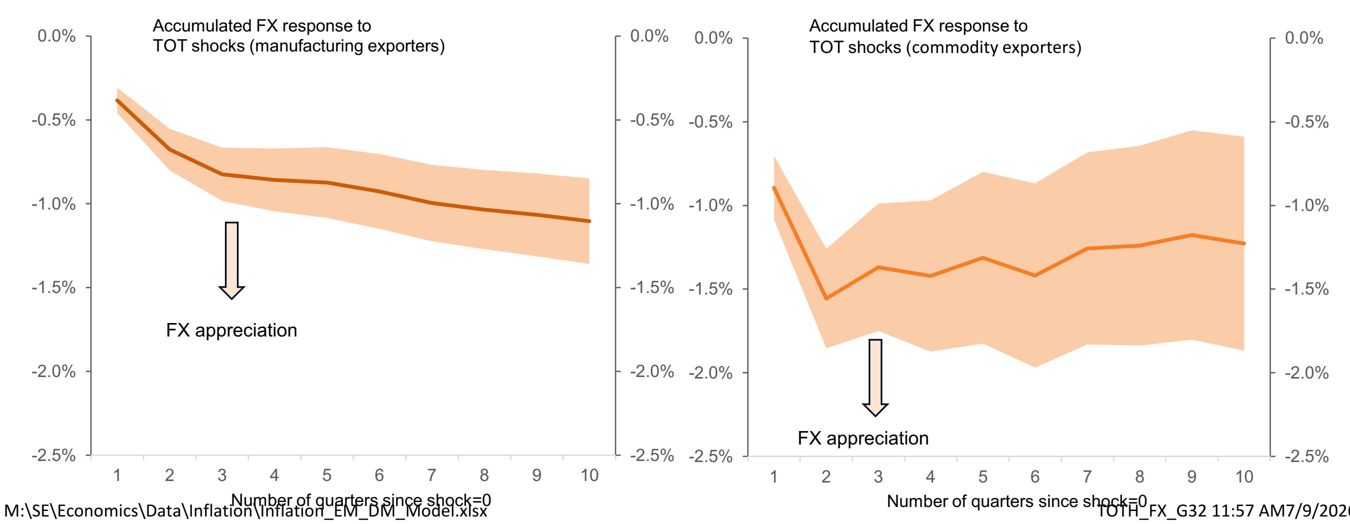
Exhibit 3: Recent terms of trade gains in commodity exporters and in Korea and Taiwan, which export semiconductors



Source: Haver Analytics, Goldman Sachs Global Investment Research

Four findings stand out in our cross-country analysis. First, terms-of-trade gains broadly strengthen currencies across exporters, in line with the findings of most empirical studies. Quarterly data suggest that a one-standard deviation gain of 2.9% in the terms of trade appreciates the currencies of manufacturing exporters by 0.5% in the near term and by 1% over four quarters. We find similar patterns in commodity exporters, although their terms of trade are three times more volatile and the transmission is faster than in manufacturing exporters (Exhibit 4).

Exhibit 4: Currencies appreciate on positive terms-of-trade shocks equal to one standard deviation both in manufacturing and commodity exporters



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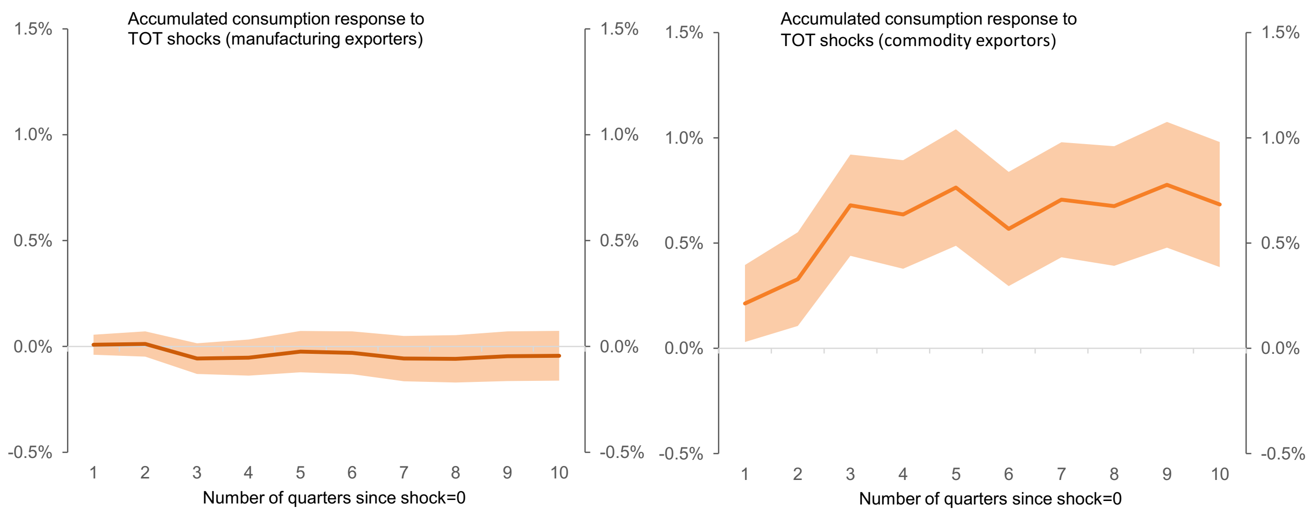
Second, the broad macro effects differ meaningfully between commodity and manufacturing exporters. Real private consumption tends to move in line with commodity prices in commodity exporters, reflecting more jobs, higher wages, and/or procyclical fiscal policy (Exhibit 5). This is consistent with the commodity boom-bust literature and raises consumption volatility as a share of GDP. In contrast, real private consumption in manufacturing exporters is largely stable and invariant to changes in the terms of trade.

Exhibit 5: Improving terms of trade tend to boost real consumption in commodity exporters but not in manufacturing exporters



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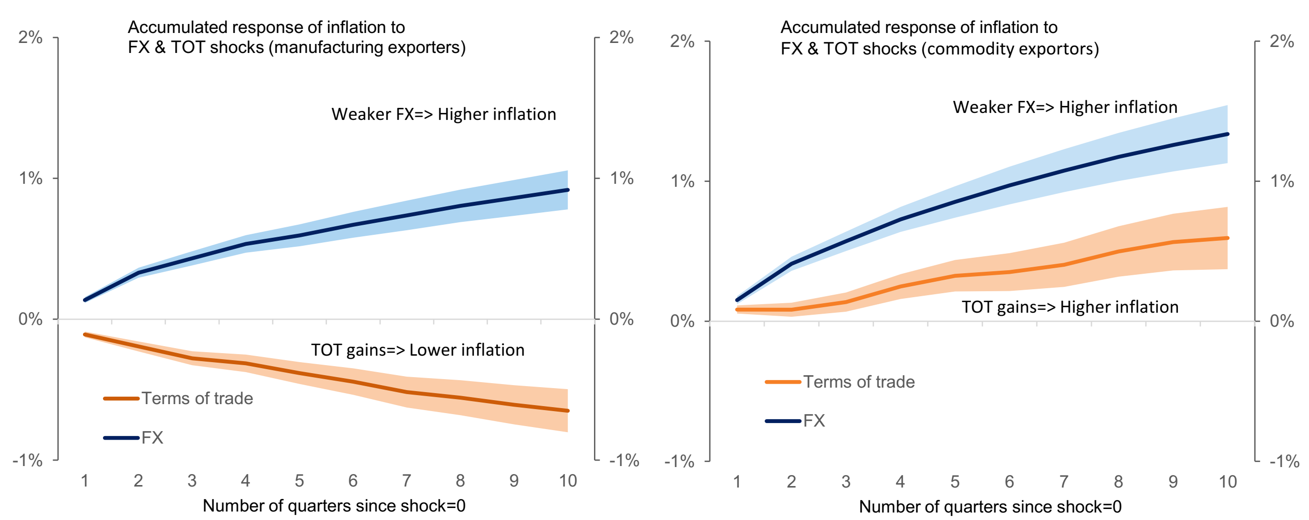
Third, these differences in currency and consumption transmission also shape inflation outcomes. Terms-of-trade gains tend to lower inflation in manufacturing exporters (Exhibit 6). The disinflationary impact of terms-of-trade gains in manufacturing exporters comes through currency appreciation, as discussed above. However, for commodity exporters, terms of trade gains increase inflation, with the disinflationary effect of stronger currencies more than offset by procyclical consumption.

Exhibit 6: Terms-of-trade gains lower inflation in manufacturing exporters but raise inflation in commodity exporters



Source: Haver Analytics, Goldman Sachs Global Investment Research

Finally, investment is another key channel through which terms-of-trade gains affect the real economy. Our panel regression, which controls for both time and country effects, suggests that a 10% rise in the terms of trade tends to increase real capex by 0.9% for manufacturing exporters (Exhibit 7). In contrast, the same terms-of-trade improvement tends to reduce capex by 0.6% for commodity exporters. This counterintuitive finding may reflect reverse causality, which is common in extractive industries with overcapacity and pricing power (such as OPEC), where lower capex tends to drive price increases and therefore terms-of-trade improvement. Currency depreciation also tends to reduce capex volumes for manufacturing exporters within the same quarter, likely reflecting higher costs of imported equipment.

Exhibit 7: Commodity exporters reduce capex on terms-of-trade gains, while capex of manufacturing exporters responds positively

Dependent variable	Real capex growth (qoq, sa, 2000-26)	
	Commodity exporters	Manufacturing exporters
Explanatory variables	Regression coefficients of a fixed effect model	
Real capex growth in the previous quarter	**-0.20**	**-0.20**
Terms of trade change (+: Improvement)	**-0.06**	**0.09**
Exchange rate (+: Depreciation)	-0.03	**-0.08**
Memorandum item:		
R-squared	26%	27%
Total observations	710	2432
Number of countries	8	24

Note: Bold coefficients are significant at a 95% confidence level.

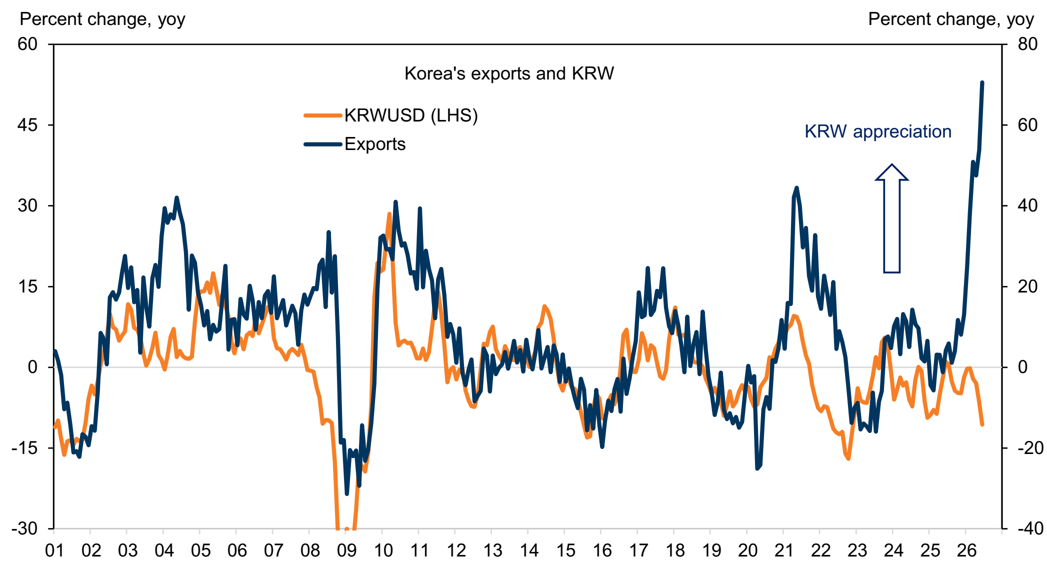
Source: Haver Analytics, Goldman Sachs Global Investment Research

Application to Korea—Similarities and Differences

Applying these cross-country patterns to Korea highlights the importance of the KRW in translating the AI boom into broader growth. The KRW has recently weakened sharply despite large terms-of-trade gains, contrary to the stylized facts (Exhibit 8). Key drivers include overseas portfolio investment by retail investors and social security funds, foreign hedging demand, and recent rebalancing by foreign investors amid the KOSPI rally.

Sustained KRW weakness could offset positive terms-of-trade spillovers to growth. In Korea, where FX passthrough to inflation is strong—about 30bp for a 5% depreciation—sustained KRW depreciation would weigh on consumers' purchasing power through higher import prices and the associated need for monetary tightening, which we expect to start at the upcoming monetary policy meeting on July 16. Profit margins of domestic companies, including local small and medium-sized suppliers to large exporters, would also decline. Moreover, ongoing structural changes in Korea's export industry reduce the positive effects of a weaker KRW on exports. We estimate that overseas sales by Korean subsidiaries abroad have increased steadily over the last decade, reaching about 90% of their direct exports from Korea onshore. In addition, more Korean companies, in particular semiconductor firms, have managed to increase their profit margins through improved productivity, reducing the significance of exchange rates on their export earnings.

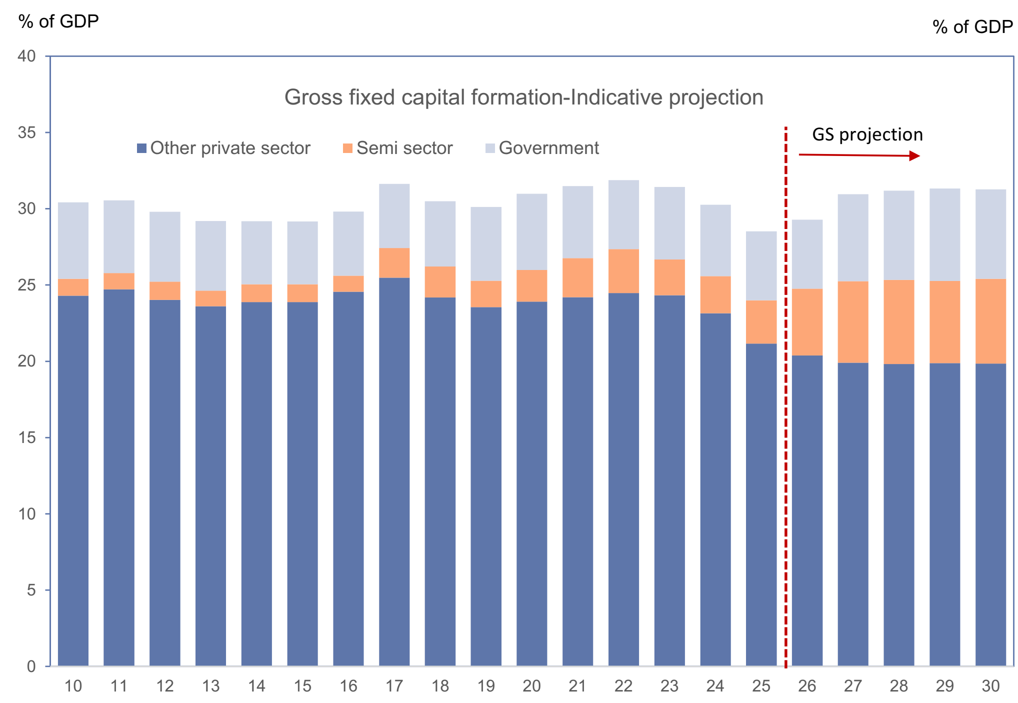
Exhibit 8: The KRW has recently become countercyclical, weakening amid surging exports



Source: Haver Analytics

While FX policy determines whether terms-of-trade gains support domestic purchasing power, fiscal policy will help determine how effectively they translate into investment. The global memory industry is currently capacity constrained, rather than facing overcapacity as in some extractive industries. Tight memory capacity bottlenecks could require substantial capex in public infrastructure as well as expanded corporate production capacity. In this context, the government has recently announced the "3 Mega Projects," a large public-private investment plan for Korea's AI development initiatives. If implemented in line with previous memory-cluster initiatives, the newly announced AI projects could increase real capex growth in Korea by 2-3pp per year and lift annual GDP growth by 0.4pp over the medium term. Under this scenario, nominal private investment in the AI projects, together with public investment, would rise by about 3-4pp of GDP over the medium term (Exhibit 9).

Exhibit 9: Mega AI projects suggest meaningful recovery in capex by the tech sector and the government



Source: Haver Analytics, Quantiwise, Goldman Sachs Global Investment Research

Policy and Market Implications

These FX and fiscal transmission channels point to two policy priorities. KRW stability would need to be a near-term policy priority to secure price stability and extend the AI boom to domestic demand. Sustained KRW weakness at the outset of the AI boom cycle would add to cost-push inflation, weigh on consumption, and widen the divergence between export and domestic industries.

Over time, however, the policy challenge is likely to shift from stabilizing prices toward recycling the surplus productively. Beyond the near term, a mix of gradual monetary tightening and saving part of the fiscal windfall—including through the creation of a sovereign wealth fund called the "Future Response Fund"—would help preserve macro stability and broaden AI-related gains across sectors. If the AI boom persists, recycling Korea's external surplus will become increasingly important for both macro stability and potential growth.

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